Supplementary Material

1. Evaluation Metrics

In this study, we use 6 classical metrics to evaluate models, namely F1-score (F1), Matthews Correlation Coefficient (MCC), Precision, Recall, the area under the receiver operator character- istic curve (AUC ROC) and the area under the precision-recall curve (AUC PR). The decision threshold is not fixed at 0.5. It is selected on the validation split by maximising F1, and the same threshold is then applied to the test set.

Precision, Recall, F1, and MCC are all threshold-dependent metrics that can be calculated as follows:

$$F1 = \frac{2 * Precision * Recall}{Precision + Recal}$$

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

where TP (True Positive) or FN (False Negative) denotes the number of binding residues that are correctly or falsely predicted. FP (False Positive) denotes the number of non-paratope residues falsely predicted as paratope. TN (True Negative) denotes the number of non-paratope residues correctly predicted as negative.

2. Supplementary Tables (S1–S3)

Supplementary Table S1. Effect of the rASA surface threshold on ParaDG performance.

Table S1 reports the prediction performance of ParaDG on the PECAN test set as the relative-solvent-accessibility (rASA) threshold varies over {15 %, 20 %, 25 %, 30 %}.

Metric	15%	20%	25% (ParaDG)	30%
AUC-ROC	0.983 (0.001)	0.983 (0.001)	**0.982 (0.001)**	**0.982 (0.001)**
AUC-PR	0.771 (0.012)	0.776 (0.012)	**0.779 (0.011)**	0.773 (0.012)
F1	0.703 (0.009)	0.704 (0.010)	**0.706 (0.009)**	0.705 (0.010)
MCC	0.691 (0.010)	0.689 (0.010)	**0.691 (0.010)**	0.689 (0.010)
Precision	**0.669 (0.011)**	0.668 (0.011)	0.640 (0.011)	0.643 (0.011)
Recall	0.748 (0.012)	0.745 (0.012)	**0.780 (0.011)**	**0.780 (0.012)**

Note. Values are reported as mean and standard deviation (in parentheses) over five independent runs with different random seeds. The adopted threshold (rASA = 25 %) reproduces the ParaDG row of the main-text Table VIII.

Supplementary Table S2. Effect of the distance metric on ParaDG oversampling.

Table S2 compares the use of Euclidean distance and cosine similarity when selecting k-nearest paratope neighbours for the structure-preserving graph oversampling module of ParaDG.

Metric	Euclidean (ParaDG)	Cosine
AUC-ROC	**0.983 (0.001)**	**0.983 (0.001)**
AUC-PR	**0.779 (0.012)**	0.778 (0.012)
F1	**0.710 (0.010)**	0.706 (0.010)
MCC	**0.696 (0.010)**	0.692 (0.010)
Precision	0.643 (0.011)	**0.655 (0.011)**
Recall	**0.792 (0.012)**	0.766 (0.013)

Note. Values are reported as mean and standard deviation (in parentheses) over five independent runs with different random seeds on the PECAN test set.

Supplementary Table S3. Effect of the *Cα–Cα* spatial-edge distance threshold on ParaDG.

Table S3 sweeps the Cα–Cα distance cutoff used in Eq. (7) over {6, 8, 10, 12, 14, 16, 18, 20} Å during antibody graph construction.

Cutoff	AUC-ROC	AUC-PR	F1	MCC	Precision	Recall
6 Å	0.981 (0.001)	0.745 (0.013)	0.701 (0.010)	0.689 (0.010)	0.621 (0.011)	**0.804 (0.011)**
8 Å	0.982 (0.001)	0.752 (0.013)	0.699 (0.010)	0.684 (0.010)	0.650 (0.012)	0.757 (0.012)
10 Å	0.983 (0.001)	0.766 (0.013)	0.709 (0.010)	0.695 (0.010)	0.653 (0.012)	0.775 (0.011)
12 Å	0.983 (0.001)	0.771 (0.012)	**0.710 (0.010)**	0.695 (0.011)	**0.684 (0.012)**	0.738 (0.013)
14 Å	0.983 (0.001)	0.774 (0.012)	0.707 (0.010)	0.694 (0.010)	0.638 (0.012)	0.793 (0.012)
16 Å **(ParaDG)**	**0.983 (0.001)**	**0.779 (0.012)**	**0.710 (0.010)**	**0.696 (0.010)**	0.643 (0.011)	0.792 (0.012)
18 Å	0.983 (0.001)	0.776 (0.012)	0.708 (0.010)	0.694 (0.010)	0.651 (0.012)	0.777 (0.012)
20 Å	0.982 (0.001)	0.768 (0.012)	0.708 (0.009)	0.695 (0.010)	0.640 (0.011)	0.791 (0.012)

Note. Values are reported as mean and standard deviation (in parentheses) over five independent runs with different random seeds on the PECAN test set.